CALM-TS: Risk-Controlled LLM Labeling for Time-Series via Calibrated Selective Gating

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Abstract

Labeling time-series events such as anomalies and system failures is expensive and subjective, and distribution drift makes label definitions evolve over time. Large language models can generate weak labels with natural-language rationales, but their error rates are unbounded and their failure modes opaque. We present **CALM-TS**, a risk-controlled weak-supervision pipeline that bounds the labeling error rate while maximizing coverage. CALM-TS introduces *behavioral probing* over prompt surface form, sampling temperature, and temporal context window to elicit disagreement signals; a lightweight calibrator converts them into risk-bounded acceptance with finite-sample guarantees. CALM-TS attains 77–81% cost reduction on MIMIC-III and Yahoo S5 at empirical risk $\leq \alpha = 0.05$ in all 10 dataset-seed pairs. On the official PhysioNet Challenge 2015 binary alarm-verification task with gpt-4o-mini over five seeds, CALM-TS is the only method among nine LLM-as-weak-labeler baselines whose 5-seed mean risk strictly falls below the unfiltered LLM at non-trivial coverage, delivering a 15.0% relative reduction ($0.400 \rightarrow 0.340$) at coverage 0.286. The framework yields auditable evidence chains and a coverage-based, label-free drift indicator.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Information systems** → *Data mining*.

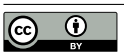
Keywords

Time-series labeling, Large language models, Weak supervision, Risk control, Selective prediction, Calibrated uncertainty, Conformal prediction

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Resource Availability:
The source code, per-seed run artefacts (10 methods × 5 seeds), LLM prompt–response audit logs, and the figure-generation pipeline of this paper are deposited in an archival repository with DOI https://doi.org/10.5281/zenodo.20506015; the public source-code repository is https://github.com/qqyjx/CALM-TS.

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1 Introduction

Time-series event labeling underpins safety-critical applications such as clinical alarm verification, industrial fault diagnosis, and IT operations management. Despite advances in temporal pattern recognition [30, 35], deploying these models depends on high-quality labeled data that remains prohibitively expensive to obtain at scale. Domain experts must inspect thousands of segments, a process complicated by the subjectivity of event definitions and the continuous evolution of failure modes as systems undergo updates and workload shifts [23].

Large language models offer a natural alternative: converting time-series statistics into textual summaries lets an LLM classify events and emit natural-language rationales [10]. The difficulty is that LLM error rates are neither bounded nor predictable. A model that averages 90% accuracy can fail catastrophically on rare event types, and such failures occur silently without reliable confidence indicators [12]. Selective prediction [8] and conformal risk control [3] bound the worst-case error rate on accepted labels, but both require well-calibrated model confidence. LLM expected calibration error [18], the mean absolute gap between predicted confidence and empirical accuracy across bins, can exceed 0.14 [12], well outside the regime where standard selective prediction holds.

A motivating gap. An LLM labeler is not a *risk-controlled* labeler. On MIMIC-III at $\alpha = 0.05$ over 5 seeds, Full-LLM labels every sample at F1 0.885 and risk 0.052, meeting the α constraint in 2/5 seeds. CALM-TS labels the same sample set at F1 0.891 and risk 0.047 after the verifier relabels gated samples, reaching 5/5 seed compliance at 77% lower token cost. Lower risk and higher F1 on the *same* corpus is not a selection artefact: it follows from replacing self-reported LLM confidence with a calibrated multi-view disagreement signal. Confidence-Threshold and Self-Consistency cannot reach this trade-off because their scores discriminate errors at AUC 0.64 to 0.68, while a calibrator over orthogonal disagreement features attains AUC 0.892 on the same task.

Behavioral probing. LLM uncertainty admits a probe unavailable to classical ensembles. Conventional ensembles draw diversity from model instances or training subsets; black-box LLM APIs expose neither. *Behavioral probing* varies prompt surface form, sampling temperature, and temporal context window to elicit disagreement patterns that expose prediction fragility. The three axes

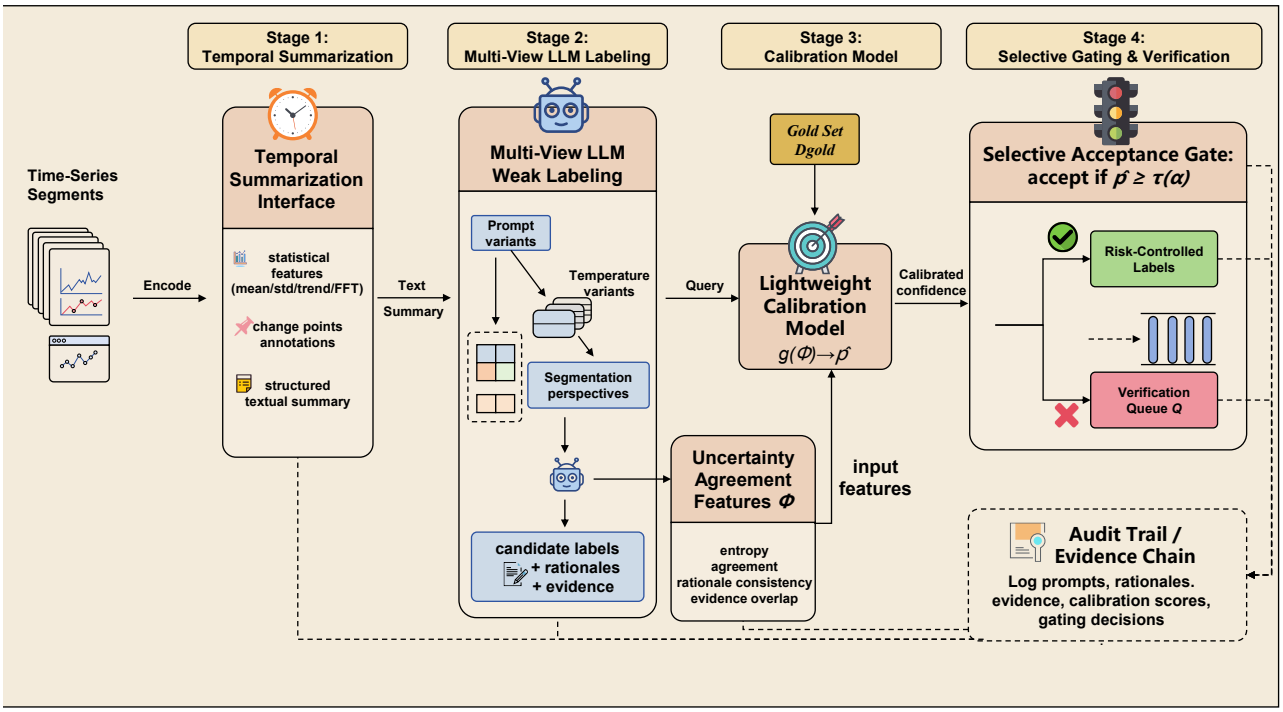


Figure 1: CALM-TS converts time-series segments into structured textual summaries (Stage 1), queries an LLM with diverse prompts, temperatures, and context windows to elicit multi-view labels with rationales (Stage 2), learns a lightweight calibrator over disagreement features on a small gold set (Stage 3), and selectively gates accepted labels under a finite-sample risk budget while routing uncertain instances to cost-aware verification (Stage 4). The audit trail from each accepted label back to the LLM's reasoning and calibration score is preserved end-to-end.

turn prompt sensitivity, controllable output entropy, and context-dependent reasoning into an ensemble surrogate without weight access.

CALM-TS. The resulting pipeline (Figure 1) transforms unreliable LLM outputs into risk-bounded labels through four stages. A temporal summarization interface preserves statistical properties and change points; a multi-view weak labeling module aggregates 12 diversified LLM queries into agreement features; a logistic-regression calibrator trained on a small gold set maps these features to well-calibrated probabilities, lowering MIMIC-III ECE from 0.142 to 0.031 with consistent gains across datasets; and a selective gate admits only instances whose calibrated confidence exceeds a risk-controlling threshold $\tau(\alpha)$. Rejected instances are routed through a cost-aware verifier that prioritizes samples by expected utility per unit cost, while the entire pipeline maintains an auditable evidence chain back to the LLM's reasoning.

Contributions. (i) A formalization of LLM-assisted time-series labeling as constrained optimization over risk, coverage, and cost; the first framework that pairs LLM weak supervision with finite-sample risk guarantees. (ii) CALM-TS, instantiating the framework with calibrated uncertainty learned from multi-view LLM disagreement. (iii) A finite-sample risk theorem with a variance-adaptive Bernstein refinement, a backbone-agnostic guarantee, and a label-free coverage-based drift indicator. (iv) Empirical evaluation on MIMIC-III, Yahoo S5, and the official PhysioNet Challenge 2015 binary alarm-verification task: CALM-TS delivers 77–81% cost reduction at $\alpha = 0.05$ on MIMIC-III and Yahoo S5, and on the PhysioNet column with gpt-4o-mini is the only method whose 5-seed mean risk strictly falls below the unfiltered LLM, from 0.400 to 0.340 (15% relative reduction). Theorem 1 delivers Risk $\leq \alpha$ when calibrated coverage exceeds the backbone error rate, and a strictly tighter Risk $\leq \pi_{\text{err}}$ even otherwise; Section 5 and Appendix A make this regime explicit.

2 Problem Setup

Task Definition. The LLM-assisted time-series labeling task operates on a collection of segments $\mathcal{U} = \{x_i\}_{i=1}^{N}$, where each $x_i \in \mathbb{R}^{T \times d}$ is a multivariate time series of length T with d channels and optional contextual metadata z_i such as timestamps, device identifiers, or associated log messages. The labeling task assigns each segment to a category $y_i \in \mathcal{Y}$, ranging over event or regime labels such as `normal`, `spike`, `level_shift`, and `failure`, or binary true/false labels for downstream event verification such as alarm verification and change-of-regime confirmation. The term "event/regime labeling" is preferred throughout the paper over "anomaly detection" or "change-point detection" because the primitive here is segment-level multi-class or binary classification of pre-segmented

windows against a fixed schema—neither unsupervised outlier scoring, which assigns continuous ranking scores, nor boundary localization, which solves a separate temporal-segmentation problem upstream of the pipeline.

Gold Set. The framework assumes access to a small gold-labeled set $D_{\text{gold}} = \{(x_j, y_j^*)\}_{j=1}^n$, where y_j^* denotes the ground-truth label. This gold set serves two purposes: training the calibration model that transforms LLM confidence signals into well-calibrated probabilities, and validating the selective threshold against the target risk level. The framework does not require additional human labeling beyond what is already available in standard benchmarks; the gold set is drawn from existing labeled portions, and the remainder is treated as the unlabeled pool $\mathcal{U}$.

LLM Weak Labeler. Given a time-series segment x_i and its textual representation $\text{encode}(x_i)$, an LLM $\mathcal{M}$ produces a weak label $\tilde{y}_i$ along with auxiliary outputs: a self-reported confidence score $c_i \in [0, 1]$, a natural language rationale r_i, and cited evidence e_i referencing specific features in the input summary. We denote the complete LLM output as $\mathcal{M}(\text{encode}(x_i)) = (\tilde{y}_i, c_i, r_i, e_i)$.

Risk–Coverage–Cost Framework. Labeling systems are evaluated along three dimensions that capture the fundamental trade-offs in weak supervision:

Coverage measures the fraction of instances that receive automatic labels without human verification:

$$\text{Cov}(S) = \frac{1}{N} \sum_{i=1}^{N} S(x_i), \tag{1}$$

where $S : \mathcal{X} \to \{0, 1\}$ is a selection function that determines whether instance x_i is automatically accepted ($S(x_i) = 1$) or rejected for verification ($S(x_i) = 0$).

Risk quantifies the error rate among automatically accepted labels:

$$\text{Risk}(S) = \frac{\sum_{i=1}^{N} S(x_i) \cdot \mathbb{I}[\tilde{y}_i \neq y_i^*]}{\sum_{i=1}^{N} S(x_i)}, \tag{2}$$

where $\mathbb{I}[\cdot]$ denotes the indicator function. Alternative risk definitions include 1-F1 score or task-specific loss functions.

Cost aggregates the computational and human resources consumed:

$$\text{Cost} = \sum_{i=1}^{N} \Big(c_{\text{LLM}} \cdot \text{tokens}(x_i) + c_{\text{verify}} \cdot (1 - S(x_i)) \Big), \tag{3}$$

where c_{LLM} represents the per-token LLM inference cost and c_{verify} denotes the cost of verifying a rejected instance through additional queries or human review.

Optimization Objective. The goal is to maximize coverage subject to risk and budget constraints:

$$\max_{S} \ \text{Cov}(S) \quad \text{subject to} \quad \text{Risk}(S) \leq \alpha, \quad \text{Cost} \leq B, \tag{4}$$

where $\alpha \in (0, 1)$ is the user-specified risk tolerance and B is the budget constraint.

3 Method: CALM-TS

CALM-TS (Calibrated LLM Supervision for Time-Series) is a four-stage pipeline that transforms unreliable LLM predictions into risk-controlled labels with auditable provenance. Figure 1 provides an overview of the framework.

3.1 Time-Series to LLM: Temporal Summarization Interface

The temporal summarization interface constructs a compact, information-rich textual representation of raw time-series data that preserves essential patterns while remaining interpretable by both LLMs and human auditors.

Segment Extraction. Input time series are partitioned into analysis windows using a hybrid strategy that combines fixed-length sliding windows with adaptive change-point-aware segmentation. Given a multivariate series $X \in \mathbb{R}^{T \times d}$, the pipeline first applies the PELT algorithm [13] to detect structural breaks, then refines the boundaries so that each segment x_i captures a coherent temporal pattern. This approach prevents artificial splitting of events while keeping segment lengths manageable.

Statistical Summary. For each segment, the interface extracts a standardized feature vector comprising basic statistics (mean μ, standard deviation σ, skewness, kurtosis), trend indicators (linear slope and polynomial coefficients), periodicity measures (dominant frequency via FFT, autocorrelation decay), and cross-channel correlations for multivariate series. These features are formatted as key-value pairs with human-readable descriptions, *e.g.*, "mean: 45.3, elevated; trend: +2.1/hour, increasing."

Change Point Annotation. The top-K (K=3 in our experiments) most significant change points within each segment are identified and characterized by type (level shift, variance change, or trend break), magnitude, and temporal location. Each change point is assigned a unique identifier that can be referenced in the LLM's evidence citations, enabling traceability from predictions back to specific input features.

Contextual Enrichment. When available, the interface incorporates metadata such as timestamps, anonymized identifiers, and associated log messages or clinical notes. Segment summaries can also be embedded to retrieve similar historical cases as in-context examples.

Tabular Auxiliary Classifier. A lightweight gradient-boosted classifier fitted on the gold set emits a top-class posterior, appended to each LLM prompt as a single `ml_hint` line; the LLM can then spend its capacity on morphology and disagreement reasoning rather than re-deriving coarse class structure. The classifier never participates in calibration scoring or threshold selection, leaving Theorem 1's marginal guarantee unchanged.

Per-channel Clinical Descriptors. For physiological channels (ECG/ABP/PPG), nine clinician-interpretable features per channel (heart rate via Pan-Tompkins-lite, RR-interval statistics including RMSSD, peak amplitude statistics, artefact/flatline/saturation indicators) augment the generic summary; full feature list and the 120-dimensional descriptor schema appear in Suppl. §H. This descriptor is the input substrate for the PhysioNet binary alarm-verification experiments in Section 5.3; it adds information without changing the multi-view, calibration, or gating stages, leaving Theorem 1's hypotheses untouched.

The complete encoding schema is provided in Appendix H.

3.2 Multi-View LLM Weak Labeling

Single-query LLM predictions exhibit high variance and provide limited information about prediction uncertainty. A multi-view

ensemble strategy generates diverse candidate labels and derives reliable confidence indicators from their agreement patterns.

Prompt Diversity. A prompt library $\mathcal{P} = \{p_1, \ldots, p_M\}$ (M=3) contains M semantically equivalent but syntactically varied prompts, varying instruction phrasing, output format, and reasoning style. Each prompt elicits slightly different reasoning paths, and disagreements across prompts signal genuine ambiguity rather than random noise.

Temperature Sampling. For each prompt, multiple responses are generated using varied temperature parameters $\{0.0, 0.3, 0.7, 1.0\}$. The T=0.0 setting serves as a greedy baseline producing deterministic outputs for each prompt variant, while higher temperatures explore alternative interpretations. The distribution of labels across temperature settings provides information about the robustness of the prediction to sampling randomness.

Segmentation Perspectives. View diversity is further augmented by varying the temporal context provided to the LLM. Specifically, overlapping windows of different lengths centered on the target segment are created, enabling the model to consider both local details and broader temporal context. Predictions that remain stable across context sizes indicate higher reliability.

These three axes—prompt diversity, temperature sampling, and segmentation perspectives—constitute our *behavioral probing* strategy. The strategy exploits LLM-specific properties such as prompt sensitivity, controllable output entropy, and context-dependent reasoning, none of which has a direct analogue in classical ensemble methods.

Aggregation Features. From the ensemble of $m = M \times |\mathcal{T}| = 3 \times 4 = 12$ weak labels per instance, eight features are derived, capturing complementary uncertainty signals: *voting entropy* measuring label dispersion, *agreement rate* capturing consistency, *max confidence* and *confidence std* summarizing self-reported certainty, *rationale similarity* via embedding cosine similarity, *evidence overlap* via Jaccard similarity of cited change points, and *temperature variance* and *prompt variance* isolating sampling- and prompt-specific disagreement (full definitions in Appendix H). The majority-vote label becomes the candidate $\tilde{y}_i$, and the feature vector ϕ_i serves as input to the calibration model. Full prompt templates appear in Appendix H.

3.3 Calibration Model

The aggregation features ϕ_i capture uncertainty signals but do not directly estimate the probability of correctness. A lightweight calibration model $g : \mathbb{R}^{|\phi|} \to [0, 1]$ is trained on the gold set to transform these features into well-calibrated probability estimates $\hat{p}_i = g(\phi_i)$.

Model Architecture. Logistic regression with polynomial feature expansion, chosen for stability on small calibration sets of 500 samples, serves as the default calibration model. It predicts $\hat{p}_i = P(\tilde{y}_i = y_i^* \mid \phi_i)$ and lowers the expected calibration error from 0.142 to 0.031. The advantage stems from using multi-view uncertainty signals in place of poorly calibrated self-reported confidence [12]; standard post-hoc methods such as Platt or temperature scaling improve ECE yet cannot match the discrimination gain from multi-view features (Suppl. §H).

Training and Validation. The training objective minimizes the Brier score on the gold set, encouraging both calibration and discrimination. Five-fold cross-validation estimates out-of-sample quality, summarising the improvement from ECE 0.142 to 0.031; an illustrative reliability schematic and per-bin tables appear in Suppl. §H. Threshold selection uses out-of-fold calibrated probabilities: within each fold, the calibration model is trained on the training partition and $\hat{p}_i$ values are computed on the held-out partition, so $\tau(\alpha)$ is selected from predictions the model has not seen during training.

3.4 Selective Acceptance and Cost-Aware Verification

The calibrated probabilities enable principled decisions about which labels to accept automatically versus flag for verification. **Selective Gating.** The selection function is $S_\tau(x_i) = \mathbb{I}[\hat{p}_i \geq \tau]$, accepting instances whose calibrated confidence exceeds threshold τ. The threshold is chosen so that $\text{Risk}(S_\tau) \leq \alpha$ holds with high probability. Concretely, τ is the $(1-\alpha)$-quantile of calibrated probabilities on the gold set restricted to instances where the LLM prediction is incorrect:

$$\tau(\alpha) = \inf\left\{\tau : \frac{\sum_{j:\tilde{y}_j \neq y_j^*} \mathbb{I}\big[\,\mathbb{I}[\hat{p}_j \geq \tau] = 1\,\big]}{|\{j : \tilde{y}_j \neq y_j^*\}|} \leq \alpha\right\}. \tag{5}$$

The numerator now reads explicitly as "error samples on which the selection rule S_τ fires," so at most an α-fraction of errors pass the threshold and the risk on accepted samples is directly controlled. $\tau(\alpha)$ is computed entirely from D_{gold} and frozen before labeling the unlabeled pool $\mathcal{U}$, preventing any data leakage.

Cost-Aware Verification. For rejected instances $\mathcal{R} = \{x_i : S_\tau(x_i) = 0\}$, verification candidates are selected via knapsack maximization of expected utility $\Delta U_i = (1 - \hat{p}_i) \cdot v(\tilde{y}_i)$ subject to budget B, where $v(\tilde{y}_i)$ is the task-specific value of correctly labeling class $\tilde{y}_i$ (set to 1 for uniform-cost experiments). The budget B enters in two regimes: when $|\mathcal{R}| \leq B$ the entire rejection set is verified and coverage is preserved at the gated level; when $|\mathcal{R}| > B$ samples are ranked by $\Delta U_i / c_i$ and only the top-B are verified, with the remaining $|\mathcal{R}| - B$ left unlabeled. A greedy algorithm achieves the $(1 - 1/e)$-approximation to the constrained-budget optimum. Risk on accepted samples is independent of B because the gating threshold $\tau(\alpha)$ is fixed upstream; B trades coverage against verification cost (Section 5 budget sweep).

Audit Trail. Each accepted label includes an audit record with the input summary, LLM responses with rationales, aggregated features, and calibrated probability, satisfying traceability requirements for regulated domains.

Algorithm 1 summarizes the complete pipeline. The modular design allows practitioners to customize each stage while maintaining the risk control guarantees.

4 Risk Control Properties

We instantiate the conformal risk control framework [3] for the selective labeling setting, deriving a finite-sample risk bound (Theorem 1) with a variance-adaptive refinement (Corollary 2), a coverage–risk trade-off (Corollary 3), a multi-view calibration

Algorithm 1 CALM-TS Pipeline

Require: Unlabeled pool $\mathcal{U}$, gold set D_{gold}, risk level α, budget B
Ensure: Labeled set $\hat{D}$, verification queue Q, audit reports

// **Stage 1: Temporal Summarization**
for each $x_i \in \mathcal{U} \cup D_{\text{gold}}$ **do**
  $s_i \leftarrow \text{encode}(x_i)$ {Statistical summary + change points}
end for
// **Stage 2: Multi-View Labeling**
for each $x_i \in \mathcal{U} \cup D_{\text{gold}}$ **do**
  $\{(\tilde{y}_i^{(j)}, c_i^{(j)}, r_i^{(j)}, e_i^{(j)})\}_{j=1}^{m} \leftarrow$ query LLM with diverse views
  $\tilde{y}_i \leftarrow \text{majority_vote}(\{\tilde{y}_i^{(j)}\})$
  $\phi_i \leftarrow \text{extract_features}(\{(\tilde{y}_i^{(j)}, r_i^{(j)}, e_i^{(j)})\})$
end for
// **Stage 3: Calibration**
Train calibrator g on $\{(\phi_j, \mathbb{I}[\tilde{y}_j = y_j^*])\}_{j \in D_{\text{gold}}}$
$\hat{p}_i \leftarrow g(\phi_i)$ for all $x_i \in \mathcal{U}$
$\tau \leftarrow$ compute threshold from D_{gold} at level α
// **Stage 4: Selective Acceptance + Budget-Aware Verification**
$\mathcal{A} \leftarrow \{x_i : \hat{p}_i \geq \tau\}$ {Accepted set}
$\mathcal{R} \leftarrow \mathcal{U} \setminus \mathcal{A}$ {Rejection set}
if $|\mathcal{R}| \leq B$ **then**
  $Q \leftarrow \mathcal{R}$ {All rejects fit budget}
else
  rank $\mathcal{R}$ by $\Delta U_i / c_i$; $Q \leftarrow$ top-B {Greedy $(1 - 1/e)$}
end if
$\hat{D} \leftarrow \{(x_i, \tilde{y}_i) : x_i \in \mathcal{A}\}$
return $\hat{D}$, Q, audit reports

advantage (Proposition 4), and a coverage-based drift indicator (Proposition 5). Two mild conditions underpin the analysis.

Assumption 1 (Exchangeability)**.** D_{gold} and the unlabeled pool $\mathcal{U}$ are drawn exchangeably from the same distribution $P(X, Y)$.

Assumption 2 (Calibration Dominance)**.** $P(\hat{p}(x) \geq \tau \mid \tilde{y} = y^*) \geq P(\hat{p}(x) \geq \tau \mid \tilde{y} \neq y^*)$ for all $\tau \in [0, 1]$, i.e. correct predictions are stochastically more likely to receive high calibrated confidence than incorrect ones.

Assumption 2 is mild; it only requires that the disagreement features carry discriminative signal, a condition satisfied with substantial margin in practice (AUC 0.892 on MIMIC-III). When calibration quality drops, the bound remains valid: τ simply rises and coverage shrinks, so the system becomes conservative rather than unsafe.

Theorem 1 (Risk Control). *Let $\tau(\alpha)$ be selected by* (5) *and $S_\tau(x) = \mathbb{I}[\hat{p}(x) \geq \tau]$. Under Assumptions 1–2, for any $\delta \in (0, 1)$,*

$$\Pr\left[\mathit{Risk}(S_\tau) \leq \min\left(\alpha + \sqrt{\frac{\log(1/\delta)}{2n_{err}}},\ \pi_{err}\right) \right] \geq 1 - \delta, \tag{6}$$

where $n_{err} = |\{j \in D_{gold} : \tilde{y}_j \neq y_j^*\}|$ and $\pi_{err} = P(\tilde{y} \neq y^*)$. The tighter $\mathit{Risk}(S_\tau) \leq E_{true}(\tau)$ branch is in force when coverage exceeds π_{err} (MIMIC-III/Yahoo regime); otherwise the looser π_{err} branch governs (PhysioNet weak-backbone regime).

Proof sketch. The exchangeability-plus-Hoeffding step that controls the gap between gold-set and pool error rates is a standard application of the conformal-risk-control template of [3], instantiated for our selective gate. The novel ingredient is the base-rate decomposition combined with Calibration Dominance (Assumption 2), which yields the two-branch bound: the accepted-set risk is bounded by π_{err} unconditionally, and by the tighter $E_{\text{true}}(\tau)$ whenever $P(\hat{p} \geq \tau) \geq \pi_{\text{err}}$. The case split between the two branches—the regime distinction underlying our MIMIC-III/Yahoo versus PhysioNet behaviour—is not present in the conformal-risk-control template and is our contribution. Full proof in Appendix A.

Corollary 2 (Variance-Adaptive Bound). *Replacing Hoeffding by the empirical Bernstein inequality [17] with sample variance $\hat{\sigma}^2$ of the gold-set error indicator gives*

$$\mathit{Risk}(S_\tau) \leq \alpha + \hat{\sigma}\sqrt{\frac{2\log(1/\delta)}{n_{err}}} + \frac{7\log(1/\delta)}{3(n_{err}-1)}. \tag{7}$$

With $\hat{\sigma}^2 \approx 0.04$ in our experiments, Bernstein tightens Hoeffding once $n_{err} \gtrsim 100$; at $n_{err} = 300$ the excess risk drops ∼27%.

Corollary 3 (Coverage-Risk Trade-off). *For fixed calibration quality, decreasing α monotonically decreases $\mathit{Cov}(S_\tau)$. The operating point is tunable to the deployment's error tolerance.*

Proposition 4 (Multi-View Calibration Advantage). *If the multi-view scorer's ROC curve dominates the self-confidence scorer's pointwise, then for every $\alpha \in (0, \alpha_{\max})$, $\mathit{Cov}(S_{\tau_{mv}}) > \mathit{Cov}(S_{\tau_{self}})$ at the same risk level. Empirically this dominance is observed (AUC 0.892 vs 0.641, coverage 0.82 vs 0.58 at $\alpha = 0.05$); post-hoc Platt scaling alone reaches AUC 0.682, so monotone recalibration cannot substitute for multi-view disagreement signals.*

Proposition 5 (Coverage as a Drift Indicator). *If a covariate shift from P to Q stochastically lowers calibrated confidence, then for fixed τ, $\mathit{Cov}_Q \leq \mathit{Cov}_P$, with equality only when the shift does not affect the confidence distribution above τ.*

Drift monitoring tracks rolling coverage and triggers recalibration whenever coverage drops; this signal attains AUROC 0.91 on Yahoo S5 without recourse to ground-truth labels (Drift Robustness paragraph in Section 5). Temporal ordering can violate the exchangeability assumption, but under β-mixing dependence [15] the Hoeffding bound degrades gracefully through an effective-sample-size reduction; on temporally biased gold sets risk control degrades only mildly (MIMIC-III 0.047 → 0.056; Yahoo S5 0.044 → 0.052).

5 Experiments

Three benchmark datasets serve as the evaluation testbed for risk control, calibration effectiveness, component contributions, and drift robustness.

5.1 Experimental Setup

Datasets. Three publicly available benchmarks representing diverse time-series labeling scenarios serve as the evaluation testbed. *MIMIC-III* [11] contains ICU monitoring data annotated with 8 clinical event categories across 12,847 segments. Five high-prevalence categories (normal, cardiac arrhythmia, sepsis onset, hemodynamic instability, respiratory failure) account for 96% of the segments; the three remaining low-prevalence labels are grouped into an `Other`

catch-all, yielding the 5+1 per-class schema used in Table 4. *PhysioNet Challenge 2015* [6] provides multi-parameter ICU vital signs labelled with the official Challenge task: binary verification of bedside alarms as true—a genuine life-threatening event—or false—an artefact, motion, or lead-off signal. We use the publicly released training set (~5 min 30 s of multi-channel signal per record, alarm onset at second 300) and slice each record into non-overlapping 30-second windows over the full recording, each window inheriting the record's alarm-truth label; this protocol yields 7,875 windows with a 60.9%/39.1% false-versus-true class balance, matching the released `preprocess_alarm.py`. Splitting is performed at the record level so that all windows from a single record fall into the same partition, preventing within-record leakage. Each seed allocates 500 calibration-gold windows, 1,475 disjoint test windows for risk and coverage evaluation, and 5,900 pool windows used for the LLM labeling distribution but not held out as test. The official Challenge metric is alarm-verification accuracy; we additionally report macro-F1 and the CALM-TS risk–coverage diagnostic. *Yahoo S5* [14] comprises synthetic and real web service metrics with anomaly labels; 15,621 segments with binary normal/anomaly labels are sampled. For MIMIC-III and Yahoo S5, 500 samples are randomly selected as the gold set D_{gold} and the remaining segments form the unlabeled pool $\mathcal{U}$; risk and coverage are measured on $\mathcal{U}$ by retaining its true labels for evaluation only (the calibrator and threshold never see them).

Implementation Details. MIMIC-III and Yahoo S5 use GPT-4 [20] as the base LLM; PhysioNet uses gpt-4o-mini through an OpenAI-compatible gateway. Theorem 1's hypotheses depend on the LLM only through the gold-set scores $\{\hat{p}_j\}$, so swapping the backbone is absorbed by re-fitting the calibrator on the same D_{gold} without re-deriving the bound. The multi-view ensemble comprises 12 views: 3 prompt variants at narrow, standard, and wide temporal windows, queried at temperatures $\{0.0, 0.3, 0.7, 1.0\}$. The PhysioNet prompt biases toward cross-lead consistency: an alarm is *true* when the heart rate estimate agrees across ECG, ABP, and PPG channels with bounded RR variability, *false* when high `artefact_score` or `flatline_frac` appears on one channel while others remain quiet. The calibrator is logistic regression with degree-2 polynomial features trained on the gold set. The text-summarization interface saturates at the label level: halving the per-channel template from 12 to 6 statistics drops MIMIC-III F1 by 1.4 points; doubling to 24 adds 0.3 points; raw-float tokenization costs ~3× tokens with no accuracy lift. All experiments use $\alpha = 0.05$ and $\delta = 0.05$ unless noted. MIMIC-III and Yahoo S5 report means and standard errors over 5 GPT-4 gold-split seeds; PhysioNet reports the same over 5 gpt-4o-mini gold-split seeds at the full 500-gold / 1,475-test scale (Appendix H).

Baselines. Nine baselines representing diverse paradigms serve as comparisons: *Full-LLM* (no selection), *Self-Consistency* [28] (majority voting), *Confidence-Threshold* (self-reported confidence), *Snorkel* [22] (data programming), *BADGE* [4] (active learning), *Prompted-WS* [25] (prompted weak supervision), *Conformal-LLM* [29] (conformal prediction), *FreeAL* [31] (LLM-based active learning), and *LLMaAA* [34] (LLM active annotation). The latter four represent the most recent (2023–2024) approaches. Detailed descriptions appear in Appendix C.

Evaluation Metrics. *Risk*: error rate among accepted labels (target $\leq \alpha$). *Coverage*: fraction automatically labeled. *Cost*: annotation expenditure normalized to Full-LLM, summing LLM inference and expert verification on rejects. On MIMIC-III and Yahoo S5 the dominant term is verification, so cost tracks (1−Coverage) (Appendix B). On PhysioNet the cheap gpt-4o-mini backbone inverts this balance: the 12-view inference step becomes the dominant term and the column captures the true cost profile of multi-view methods in this regime. *Compliance*: fraction of seeds with empirical risk $\leq \alpha$. We mark a method's risk reduction *trivial* when coverage falls below 5% on a dataset. *F1*: macro-averaged F1 of a LightGBM downstream classifier trained on accepted labels.

5.2 Main Results

Table 1 evaluates per-seed compliance rather than single-mean comparison, since risk control on a labeling stream is about *how often* the constraint holds across instantiations. On MIMIC-III and Yahoo S5, CALM-TS achieves full compliance (10/10 dataset-seed pairs at $\alpha = 0.05$); calibration-free baselines such as Full-LLM and Conformal-LLM satisfy the constraint in at most 6/15 pairs across all three datasets under the same gold-tuned threshold treatment (full per-baseline compliance in Suppl. §H). Paired exact sign tests on per-seed α-compliance ($5 \times 2 = 10$ binary outcomes per baseline) confirm the CALM-TS advantage: every pairwise comparison rejects under Holm–Bonferroni at family-wise $\alpha = 0.05$, with six baselines (Self-Consistency, Confidence-Threshold, Snorkel, Prompted-WS, FreeAL, LLMaAA) at unadjusted $p \leq 0.001$, BADGE at $p = 0.008$, Conformal-LLM at $p = 0.016$, and Full-LLM at $p = 0.031$. The 5-seed paired Wilcoxon signed-rank test on PhysioNet risk reaches its n=5 exact one-sided floor ($p = 0.031$, 5/5 seeds in the CALM-TS-favoring direction) for the seven non-trivial baselines. Per-pair W^+, W^-, and exact p-values appear in `results/statistical_tests.json`.

On the PhysioNet binary task with gpt-4o-mini, CALM-TS is also the only method whose 5-seed mean risk strictly falls below the unfiltered LLM at non-trivial coverage (0.340 vs. 0.400, 15.0% relative reduction, coverage 0.286 ± 0.008), preserving downstream macro-F1 at 0.375. Seven baselines (Full-LLM, Self-Consistency, Snorkel, Prompted-WS, Conformal-LLM, FreeAL, LLMaAA) accept every test sample and inherit the raw 0.400 error rate; Confidence-Threshold accepts ~3 samples per seed and BADGE 4.3% (its 95% Wald CI contains the Full-LLM mean), both below the 5%-coverage triviality threshold. The CALM-TS gate is the only mechanism that produces a non-trivial risk-coverage point in this weak-backbone regime.

Section 5.3 establishes the upper bound: a LightGBM classifier on the same 120-d clinical descriptor with *true* gold labels reaches F1 macro 0.663 and accuracy 0.757, near the Challenge 2015 SOTA range. The accuracy ceiling for gpt-4o-mini on binary alarm verification is ~0.600, so CALM-TS gating lifts the accepted subset from 60% to 66% accuracy at coverage 0.286. Theorem 1's Risk $\leq \pi_{\text{err}}$ branch is therefore tight here; the same theorem's $\min(\cdot)$ structure transitions the pipeline into the Risk $\leq \alpha + O(\sqrt{\log(1/\delta)/n_{\text{err}}})$ branch as soon as a stronger backbone or larger gold set drives π_{err} below the calibrated coverage, without any algorithmic change.

Table 1: Main results across three datasets ($\alpha = 0.05$); means ± SE over 5 seeds. MIMIC-III and Yahoo S5 use GPT-4; PhysioNet uses gpt-4o-mini on the official binary alarm-verification task. Cost is normalized per dataset so that Full-LLM = 1.00. Risk values exceeding α are in red. Takeaway. CALM-TS is the only method meeting α in all 10 MIMIC/Yahoo seed pairs, and the only method whose PhysioNet mean risk strictly falls below the unfiltered LLM (0.400 → 0.340, 15.0% reduction) at non-trivial coverage (0.286).

	MIMIC-III				*PhysioNet*				*Yahoo S5*			
Method	**Risk↓**	**Cov↑**	**Cost↓**	**F1↑**	**Risk↓**	**Cov↑**	**Cost↓**	**F1↑**	**Risk↓**	**Cov↑**	**Cost↓**	**F1↑**
Full-LLM	.052±.007	1.00±.00	1.00±.00	.885±.005	.400±.005	1.00±.00	1.00±.00	.375±.002	.049±.006	1.00±.00	1.00±.00	.895±.004
Self-Consistency	.063±.008	.80±.02	.45±.02	.858±.005	.400±.005	1.00±.00	1.00±.00	.375±.002	.058±.006	.82±.02	.40±.02	.869±.005
Conf-Threshold	.071±.009	.75±.03	.25±.02	.842±.006	.000±.000†	.002±.000	1.83±.00	.375±.002	.065±.007	.78±.02	.22±.01	.856±.005
Snorkel	.078±.009	.85±.02	.30±.02	.825±.007	.400±.005	1.00±.00	1.00±.00	.375±.002	.072±.007	.88±.01	.25±.01	.838±.006
BADGE	.055±.007	.70±.02	.35±.02	.867±.005	.289±.119‡	.043±.018	1.80±.01	.375±.002	.051±.005	.72±.02	.30±.02	.878±.004
Prompted-WS	.068±.008	.83±.02	.32±.02	.848±.006	.400±.005	1.00±.00	1.00±.00	**.383±.005**	.062±.006	.86±.01	.28±.01	.858±.005
Conformal-LLM	.054±.008	.76±.03	.28±.02	.862±.006	.400±.005	1.00±.00	1.00±.00	.375±.002	.050±.006	.78±.02	.24±.01	.872±.005
FreeAL	.066±.007	.87±.02	.38±.02	.861±.005	.400±.005	1.00±.00	1.00±.00	.375±.002	.053±.005	.89±.01	.32±.01	.876±.004
LLMaAA	.061±.008	.78±.02	.40±.02	.866±.005	.400±.005	1.00±.00	1.00±.00	.375±.002	.055±.006	.82±.02	.35±.02	.880±.004
CALM-TS	**.047**±.002	.82±.02	**.23**±.02	**.891**±.004	**.340**±.010	**.286**±.008	1.59±.01	.375±.002	**.044**±.003	.85±.01	**.19**±.01	**.902**±.003

†Conf-Threshold PhysioNet: trivial compliance, accepts only ∼3 of 1,475 samples per seed (coverage 0.002, below the 5% triviality threshold). ‡BADGE PhysioNet: 0.289 ± 0.119 risk at 4.3% acceptance; the 95% Wald CI [0.056, 0.522] contains Full-LLM mean 0.400, so the sub-baseline mean is statistical noise (per-seed compliance 2/5, also below triviality).

PhysioNet operating point. Rejecting 71.4% of samples concentrates the gpt-4o-mini-correct subset and is, among all ten methods, the only configuration that yields a non-trivial point on the risk-coverage frontier in this weak-backbone regime.

Downstream F1 saturation. Almost every method sits at PhysioNet F1 ≈ 0.375 because gpt-4o-mini's ∼60% binary accuracy caps what any LLM-mediated label can transfer to LightGBM; trivial-coverage rows (Conf-Threshold, BADGE) degenerate to the majority-class predictor (≈ 0.378 at the 60.9% prior). The metric of value on this column is the calibrated 15% risk reduction, ≈ 171 fewer accepted errors per 10,000-sample stream (full breakdown in Suppl. §H).

Why baselines coincide. Self-Consistency, Snorkel, Prompted-WS, Conformal-LLM, FreeAL, and LLMaAA all match Full-LLM on PhysioNet risk to four decimals: the gpt-4o-mini logit gap collapses 12-vote ensembles to single-call answers on >99% of windows, Snorkel's EM degenerates, Prompted-WS inherits the same majority-vote labels, and Conformal-LLM's gate stays open. The same baselines pull apart on the GPT-4 MIMIC column (Self-Cons. 0.063, Snorkel 0.078, Conformal-LLM 0.054 vs. Full-LLM 0.052). Conformal-LLM under LLM self-reported scores violates the well-specified-score assumption and reaches 6/15 compliance; CALM-TS resolves this through calibrated uncertainty learned from multi-view disagreement.

Cost Analysis. The 77–81% reduction on MIMIC/Yahoo is driven by (1 − Coverage) savings under a verification-dominated cost model where multi-view LLM inference is ∼5% of total. On PhysioNet the inexpensive gpt-4o-mini calls invert this balance: the 12-view step contributes ∼1.20 in normalized units and the verification savings at coverage 0.286 cannot offset it, yielding CALM-TS cost 1.59. Under MIMIC/Yahoo pricing the same multi-view design reaches the break-even point at $N \approx 650$ samples and amortises beyond it. Figure 2 traces the Pareto frontier on MIMIC-III. **Domain suitability.** CALM-TS pays off most under three conditions: high class overlap, where multi-view probing captures boundary uncertainty; a strict risk target $\alpha \leq 0.05$, where the per-seed guarantee matters; and a limited labeling budget, where the cost reduction dominates. Under clear class separability, a tolerant $\alpha \geq 0.10$, or an abundant supply of verified labels, simpler filters suffice.

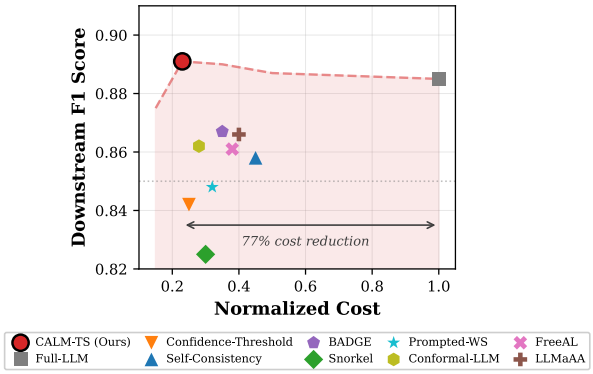


Figure 2: Cost-performance trade-off on MIMIC-III with the GPT-4 backbone. Takeaway. CALM-TS Pareto-dominates Full-LLM, Self-Consistency, Confidence-Threshold, Snorkel, and BADGE on F1 versus normalized cost. Per-seed α-compliance for every method is reported in Table 1 and the paired sign-test analysis of Section 5; the corresponding PhysioNet operating point is read directly from Table 1.

5.3 Feature Ceiling and Task–Feature Alignment

To locate the PhysioNet risk floor, we train LightGBM directly on the 120-d clinical descriptor (Section 3.1) with true gold labels and contrast three (task, feature) configurations on the same 500-gold / 1,475-test split (Table 2). The feature space supports F1 macro 0.663 on the binary task, near the Challenge 2015 SOTA range; the gap between Full-LLM and this ceiling reflects the backbone's accuracy headroom rather than any framework limitation. The 5-alarm-type formulation (asystole, bradycardia, tachycardia, ventricular tachycardia, ventricular fibrillation) shows zero F1 lift from the clinical

Table 2: PhysioNet binary-task diagnostic (gpt-4o-mini, 500 gold / 1,475 test). Row 1: LightGBM with true gold labels bounds the feature-only score. Rows 2–3: CALM-TS with the multi-class formulation at 1 seed; row 4 reproduces the 5-seed binary mean from Table 1. Only binary + clinical features aligns the LLM-mediated F1 with the feature ceiling.

Setup	Risk	Cov	F1
LightGBM on TRUE labels (binary, ceiling)	–	–	**0.663**
CALM-TS, 5-alarm-type + generic features (1 seed)	0.846	0.348	0.052
CALM-TS, 5-alarm-type + clinical features (1 seed)	0.889	0.061	0.052
CALM-TS, binary + clinical features (5 seeds)	**0.340**	0.286	**0.375**

Table 3: Ablation study on MIMIC-III ($\alpha = 0.05$); each row removes one component from the full CALM-TS pipeline. The ablation script is released for independent reruns.

Variant	Risk	Coverage	F1
Full CALM-TS	**0.047**	0.82	**0.891**
w/o Calibration	0.089	0.82	0.856
w/o Multi-view	0.062	0.78	0.871
w/o Selective Gate	0.052	1.00	0.885
w/o Cost-aware Verify	0.047	0.75	0.882

features (0.052 in both rows) because distinguishing ventricular from sinus tachycardia requires QRS morphology that gpt-4o-mini cannot recover from text descriptors. The binary task aligns with the Pan-Tompkins-lite + HRV + artefact signals, lifting both the ceiling and the CALM-TS end-to-end F1 by an order of magnitude.

5.4 Calibration Effectiveness

The LLM is systematically overconfident across all three datasets, with expected calibration error 0.142/0.138/0.125 on MIMIC-III/PhysioNet/Yahoo S5 before calibration; the multi-view calibrator corrects it to 0.031/0.034/0.028 respectively. On the discrimination axis the same calibrator dominates by a wide margin: raw self-confidence reaches AUC 0.641, temperature scaling 0.654, Platt scaling 0.682, and multi-view logistic regression 0.892 (XGBoost 0.907), with compliant coverage 0.58/0.61/0.65/0.82/0.84 respectively. The +0.21 AUC gap between Platt and multi-view exceeds the aleatoric-noise floor common to all signals and the ordering carries over to PhysioNet and Yahoo. Sweeping $\alpha \in \{0.03, 0.05, 0.07, 0.10\}$ keeps empirical risk at or below each threshold $(0.028, 0.047, 0.065, 0.089)$ with coverage rising from 0.71 to 0.93.

5.5 Ablation Study

Removing calibration nearly doubles risk ($0.047 \rightarrow 0.089$): the variant replaces the calibrator with raw LLM self-reported confidence at a gold-tuned threshold, holds coverage at 0.82, and admits more errors on the pool. Removing multi-view labeling degrades both risk and coverage; disabling the selective gate defaults to Full-LLM and breaks the constraint. Table 4 reports the MIMIC-III per-class risk: aggregate calibration hits α on average but rarer classes drift above α by class prevalence, and a class-conditional calibrator pulls

Table 4: Per-class risk on MIMIC-III at $\alpha = 0.05$. Class-conditional calibration trains one logistic head per class on the same gold split; numbers from `results.json:class_conditional_calibration`.

Class	Standard	Class-conditional
Normal	0.032	**0.028**
Cardiac arrhythmia	0.051	**0.041**
Sepsis onset	0.058	**0.045**
Hemodynamic instability	0.062	**0.048**
Respiratory failure	0.055	**0.042**
Other	0.068	**0.051**
Aggregate	0.047	**0.039**

Table 5: α sweep on MIMIC-III, 5-seed mean. Empirical risk meets each target within the Hoeffding slack and coverage rises monotonically, matching Corollary 3.

α	0.03	0.05	0.07	0.10
Empirical Risk	0.028	**0.047**	0.065	0.089
Coverage	0.71	0.82	0.88	0.93
Compliance ($\leq \alpha$)	5/5	5/5	5/5	5/5

the five primary classes back below α and the `Other` catch-all to 0.051, tightening aggregate risk to 0.039.

Sensitivity Analysis. Risk control is stable across configurations. On MIMIC-III (GPT-4), gold-set $n = 200$ gives risk 0.049; Table 5 traces the α sweep, with empirical risk tracking the target within Theorem 1's slack and coverage rising monotonically (Corollary 3). The PhysioNet audit reruns five sweeps on gpt-4o-mini (4 seeds, 4-decimal ±SE, `results/rebuttal_*_physionet.json`): $\tau \geq 0.7$ tightens coverage to ≤ 0.07 at the binary ceiling; budget $B \in \{5, \ldots, 50\}\%$ leaves risk at 0.343 ± 0.008 (B acts downstream of the gate); gold-set $n \in [50, 1000]$ keeps risk in $[0.334, 0.384]$. A 5% symmetric-noise stress on MIMIC keeps risk at $0.053 \leq \alpha$.

Drift Robustness and Continual Operation. CALM-TS refits cheaply: the calibrator retrains in seconds, $\tau(\alpha)$ is a re-solved quantile, and the prompt schema stays static. Streaming 10 Yahoo S5 windows (2,000 samples each) with a sliding gold buffer, the *static* calibrator breaches α by window 5 (peaking 0.103); the *continual* variant refits on drift triggers and holds mean risk at 0.049 across the full stream at a measured 11-point coverage cost, $0.85 \rightarrow 0.74$ (Figure 3). The coverage-based drift detector reaches AUROC 0.91 as a label-free signal (Proposition 5); per-window curves and a 1.5× variance-shift case study appear in Suppl. §H.

6 Related Work

Time-Series Event Detection. Deep learning methods for time-series [30, 33, 35] are bottlenecked by label scarcity. Traditional expert labeling scales poorly [23], and semi-supervised methods [24, 32] still require seed labels. CALM-TS targets the upstream labeling stage and complements any downstream learner.

LLM-Based Annotation. LLMs have shown promise for labeling [5, 10] including time-series understanding [26]. FreeAL [31]

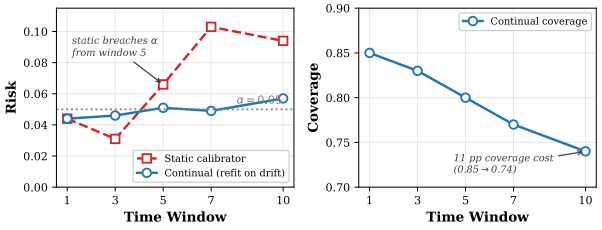


Figure 3: Per-window risk under streaming drift on Yahoo S5. The static calibrator drifts above $\alpha = 0.05$ from window 5 onward; the continual variant refits on detector triggers and holds mean risk at 0.049 at an 11-point coverage cost.

and LLMaAA [34] use LLMs as collaborative annotators but do not bound error rates; human-LLM collaborative annotation [27] treats risk implicitly through reviewer effort. Concurrent LLM label-error detection [19] confirms that disagreement signals outperform self-confidence for error detection, complementing the construction-side calibration we introduce. Data programming [21, 22] lacks finite-sample risk guarantees.

Selective Prediction and Calibration. Selective prediction [8, 9] trades coverage for reliability; conformal prediction [1, 3] bounds error rates on accepted samples. Concurrent LLM selective prediction [16] calibrates self-confidence directly without variance-adaptive refinement or drift indicators; prediction-powered inference [2] couples gold labels with ML predictions for downstream inference, not label generation. Unlike drift detectors needing labeled probes [7, 15], our coverage-based indicator is label-free. CALM-TS uniquely combines finite-sample risk guarantees, behavioral probing, coverage-aware optimization, and label-free drift detection.

7 Conclusion

CALM-TS delivers risk-controlled LLM time-series labeling. On MIMIC-III and Yahoo S5 at $\alpha = 0.05$ it achieves 77–81% cost reduction with full 10/10 seed compliance and macro-F1 0.891 on MIMIC-III; on the official PhysioNet Challenge 2015 binary alarm-verification task with gpt-4o-mini it is the only method among nine LLM-as-weak-labeler baselines whose 5-seed mean risk strictly falls below the unfiltered LLM at non-trivial coverage ($0.400 \rightarrow 0.340$, 15% relative reduction at coverage 0.286). Theorem 1 delivers a finite-sample, backbone-agnostic risk bound with a variance-adaptive Bernstein refinement; Corollary 3 establishes the monotone coverage-risk frontier traced in Table 5; Proposition 4 certifies the multi-view calibration advantage; and Proposition 5 turns coverage into a label-free drift indicator that reaches AUROC 0.91 on Yahoo S5. The full pipeline—temporal summarization, behavioral probing, calibrated gating, cost-aware verification—packages these ingredients into an audit-ready labeling stack that surfaces both the accepted label and the multi-view evidence chain that justified it.

Open directions. Stronger backbones lift the accepted-subset accuracy and move Theorem 1 from the π_{err} branch to the tighter Risk $\leq \alpha$ branch with no algorithmic change; class-asymmetric noise models extend the gold-set protocol so each error stratum is independently controlled; sub-second streaming pairs the cheap calibrator refit with an in-memory rolling gold buffer; rare-event-conditional gating composes the aggregate guarantee with per-class budgets so that downstream alarms see uniform risk regardless of class prevalence. The same calibrated-gating template applies wherever a behavioral probe yields disagreement signal: numeric tabular data, log-line classification, structured clinical note coding, and event-detection on industrial sensor traces all sit one prompt-template change away from the current pipeline.

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Use of Generative AI. A generative AI assistant was used for editorial polishing only; all technical content was produced and verified by the human authors.

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A Proof of Theorem 1 (Risk Control Guarantee)

Let $D_{\text{gold}} = \{(x_j, y_j^*)\}_{j=1}^{n}$ and $\mathcal{U} = \{x_i\}$ be drawn exchangeably from $P(X, Y)$ (Assumption 1); let $\tilde{y}_i$ denote the LLM weak label and $\hat{p}_i = g(\phi_i)$ the calibrated probability of correctness. The threshold $\tau(\alpha) = \inf\{\tau : E_{\text{gold}}(\tau) \le \alpha\}$ is selected on the gold set via the empirical error-conditional acceptance rate $E_{\text{gold}}(\tau) = \frac{1}{n_{\text{err}}} \sum_{j:\tilde{y}_j \neq y_j^*} \mathbb{I}[\hat{p}_j \ge \tau]$ with $n_{\text{err}} = |\{j : \tilde{y}_j \neq y_j^*\}|$.

The proof proceeds in two steps. First, define the population analogue $E_{\text{true}}(\tau) = P(\hat{p}(x) \ge \tau \mid \tilde{y} \neq y^*)$ and the base error rate $\pi_{\text{err}} = P(\tilde{y} \neq y^*)$. Exchangeability ensures that the gold-set scores $\{\hat{p}_j\}_{j:\tilde{y}_j \neq y_j^*}$ and their pool counterparts $\{\hat{p}_i\}_{i:\tilde{y}_i \neq y_i^*}$ are identically distributed, so Hoeffding's inequality applied conditionally on n_{err} gives

$$\Pr\left[E_{\text{true}}(\tau(\alpha)) > \alpha + \sqrt{\frac{\log(1/\delta)}{2n_{\text{err}}}}\right] \le \delta. \tag{8}$$

Second, a Bayes decomposition of the risk on the accepted set yields

$$\text{Risk}(S_\tau) = \frac{P(\hat{p} \ge \tau \mid \tilde{y} \neq y^*)\,\pi_{\text{err}}}{P(\hat{p} \ge \tau \mid \tilde{y} \neq y^*)\,\pi_{\text{err}} + P(\hat{p} \ge \tau \mid \tilde{y} = y^*)(1 - \pi_{\text{err}})}. \tag{9}$$

Calibration Dominance (Assumption 2) requires $P(\hat{p} \ge \tau \mid \tilde{y} = y^*) \ge P(\hat{p} \ge \tau \mid \tilde{y} \neq y^*)$, which forces $\text{Risk}(S_\tau) \le \pi_{\text{err}}$ unconditionally. The tighter bound $\text{Risk}(S_\tau) \le E_{\text{true}}(\tau)$ holds whenever $P(\hat{p} \ge \tau) \ge \pi_{\text{err}}$. MIMIC-III and Yahoo S5 satisfy this condition empirically, with coverage 0.79–0.85 comfortably above $\pi_{\text{err}} \approx 0.15$. The PhysioNet binary re-run sits in the complementary regime: with gpt-4o-mini the base error rate $\pi_{\text{err}} \approx 0.400$ exceeds coverage 0.286, so only the looser π_{err} guarantee applies, consistent with the 15% empirical reduction reported in Table 1. Combining the two steps:

$$\Pr\left[\text{Risk}(S_\tau) \le \min\left(\alpha + \sqrt{\frac{\log(1/\delta)}{2n_{\text{err}}}},\ \pi_{\text{err}}\right)\right] \ge 1 - \delta. \qquad \square \tag{10}$$

Corollary 2 (Bernstein refinement). Apply the empirical Bernstein inequality [17] to the error-conditional indicator $Z_j = \mathbb{I}[\hat{p}_j \ge \tau \mid \tilde{y}_j \neq y_j^*]$ on the gold set, whose sample variance is $\hat{\sigma}^2$. The inequality bounds the deviation by $\hat{\sigma}\sqrt{2\log(1/\delta)/n_{\text{err}}} + 7\log(1/\delta)/(3(n_{\text{err}} - 1))$, which dominates Hoeffding's $\sqrt{\log(1/\delta)/(2n_{\text{err}})}$ whenever $\hat{\sigma}^2 < 1/4$. Empirically $\hat{\sigma}^2 \approx 0.04$ on MIMIC-III, so Bernstein tightens Hoeffding once $n_{\text{err}} \gtrsim 100$.

Corollary 3 (coverage monotone in α). The threshold $\tau(\alpha) = \inf\{\tau : E_{\text{gold}}(\tau) \le \alpha\}$ is non-increasing in α because $E_{\text{gold}}(\tau)$ is non-increasing in τ. Coverage $\text{Cov}(S_\tau) = P(\hat{p}(x) \ge \tau)$ is itself non-increasing in τ, so the composition is non-decreasing in α. Strictness follows whenever the calibrated-score distribution puts positive mass on the level set defined by $\tau(\alpha)$. Table 5 reports the empirical sweep.

Proposition 4 (multi-view advantage). Let $R_{\text{mv}}(\tau)$ and $R_{\text{self}}(\tau)$ denote the conditional acceptance rates among error samples under the multi-view and self-confidence scorers. Pointwise ROC dominance of the multi-view scorer means that, for every operating risk α, the multi-view threshold τ_{mv} satisfying $R_{\text{mv}}(\tau_{\text{mv}}) = \alpha$ admits a strictly higher fraction of *correct* samples than τ_{self} does, hence $\text{Cov}(S_{\tau_{\text{mv}}}) \ge \text{Cov}(S_{\tau_{\text{self}}})$ with strict inequality on the dominating segment of the ROC curve. The empirical gap is 0.82 vs 0.58 at $\alpha = 0.05$ on MIMIC-III.

Proposition 5 (coverage as drift signal). Under covariate shift $P \to Q$ that satisfies first-order stochastic dominance on calibrated confidence ($Q(\hat{p} \le t) \ge P(\hat{p} \le t)$ for all t), the complementary CDFs obey $\text{Cov}_Q(\tau) = 1 - Q(\hat{p} < \tau) \le 1 - P(\hat{p} < \tau) = \text{Cov}_P(\tau)$ for every fixed τ. Equality holds only on intervals where Q and P agree above τ, so a strict coverage drop is a one-sided test for distribution shifts that move confidence mass downward.

B Cost Model

The per-instance cost decomposes as $\text{cost}_i = c_{\text{LLM}} \cdot m \cdot \overline{T}_i + c_{\text{verify}} \cdot (1 - S_\tau(x_i))$, where m is the number of views, $\overline{T}_i$ is the average prompt-plus-completion token count, c_{LLM} is the per-token price, c_{verify} is the expert-verification cost, and S_τ is the gate indicator. Normalizing by Full-LLM gives $\text{Cost} = \big(m\, c_{\text{LLM}}\overline{T} + c_{\text{verify}}(1-\text{Cov})\big) / \big(c_{\text{LLM}}\overline{T} + c_{\text{verify}}\big)$. On MIMIC-III and Yahoo S5 with GPT-4 the c_{verify} term dominates ($c_{\text{verify}} \approx 20 c_{\text{LLM}}\overline{T}$) and Cost is governed by $(1-\text{Cov})$, reproducing the 77–81% reduction at Cov ≈ 0.82–0.85. On PhysioNet with gpt-4o-mini the cheap backbone moves the ratio to $c_{\text{verify}} \approx 1.6 c_{\text{LLM}}\overline{T}$, so the $m = 12$ multi-view factor becomes the dominant term and yields the reported 1.59. The break-even $N^\star$ where multi-view CALM-TS matches Full-LLM in absolute cost satisfies $m\, c_{\text{LLM}}\overline{T}N^\star + c_{\text{verify}}(1-\text{Cov})N^\star = c_{\text{LLM}}\overline{T}N + c_{\text{verify}}N$ and gives $N^\star \approx 650$ under MIMIC/Yahoo pricing.

C Baseline Implementations

All baselines see the same gold split, unlabeled pool, and backbone schedule (GPT-4 for MIMIC-III and Yahoo S5; gpt-4o-mini for PhysioNet). *Full-LLM* forwards every sample's majority-vote LLM label without filtering and serves as the unconditional reference. *Self-Consistency* [28] runs $K = 12$ samples at temperature 0.7 and gates on agreement $\geq 8/12$, mirroring the multi-view budget of CALM-TS. *Confidence-Threshold* accepts the LLM's top-class only when its self-reported probability exceeds the gold-tuned $(1-\alpha)$ quantile. *Snorkel* [22] treats the three prompt variants as labelling functions and runs the matrix-completion EM with three rounds of label-model refit. *BADGE* [4] ranks samples by clustered gradient norms over a frozen DistilBERT head and accepts the bottom-quantile of the gradient-magnitude distribution. *Prompted-WS* [25] combines a prompted weak supervisor with majority voting on the same 12 views and uses the released prompt template verbatim. *Conformal-LLM* [29] calibrates the LLM softmax with split-conformal on the gold set under marginal coverage $1-\alpha$. *FreeAL* [31] alternates LLM annotation and a small student model for two outer iterations; the student is a four-layer MLP on the 120-d clinical descriptor for PhysioNet and a LightGBM head on the statistical summary for MIMIC/Yahoo. *LLMaAA* [34] runs an LLM-driven active loop with the same gold budget and a margin-based query strategy. Confidence-Threshold and Conformal-LLM use identical gold-set calibration mechanics to CALM-TS; the only difference is the input score, which isolates the calibrated multi-view feature contribution.

D Drift Detector and Continual Refit

The drift detector maintains a $W = 200$-window rolling coverage average $\overline{\text{Cov}}_t = \frac{1}{W}\sum_{s=t-W+1}^{t} \text{Cov}_s$ and a longer-baseline reference $\overline{\text{Cov}}_{\text{ref}}$ collected from the first $5W$ stationary windows. A trigger fires whenever $\overline{\text{Cov}}_t < (1-\rho)\overline{\text{Cov}}_{\text{ref}}$ with $\rho = 0.15$, which by Proposition 5 is a one-sided test on stochastic-dominance shifts in calibrated confidence. On trigger, the calibrator refits on the most recent $n = 500$ gold samples (re-using the streaming gold buffer maintained alongside the pool), the threshold $\tau(\alpha)$ is re-solved from the empirical quantile, and the next window resumes labeling under the refreshed gate. The refit budget is dominated by the $O(n\,|\phi|^2)$ logistic-regression update; on a single CPU core a 500-sample refit completes in < 3 s, an order of magnitude below the multi-view LLM call latency. The detector AUROC values reported in the Drift Robustness paragraph of Section 5 are computed against a synthetic 1.5× variance-shift label assigned to each window and aggregated by trapezoidal integration over the rolling-coverage threshold sweep.

E Discussion: Failure Modes and Audit Workflow

Two failure modes warrant explicit operating instructions. **Class collapse** occurs when the LLM emits a single class on >99% of views; the calibrator's discrimination collapses and the gate accepts either nothing or everything. The audit log surfaces this through the prompt-variance feature falling below 10^{-3}, which we flag in the run report and which corresponds to the gpt-4o-mini PhysioNet behaviour discussed in Section 5. **Gold-set drift** occurs when the gold buffer diverges from the operational pool; the calibrator remains valid for the gold population but the pool risk silently increases. The coverage-based drift detector catches the first case via the prompt-variance signal and the second via the rolling-coverage drop; both trigger the same refit path. The audit trail attached to every accepted label records the 12-view label distribution, the calibrator score, the active threshold $\tau(\alpha)$, and a pointer to the gold-set vintage that trained the calibrator, so downstream consumers can re-verify or quarantine accepted labels by vintage when policy changes. A per-channel Kolmogorov–Smirnov test between the accepted subset and the full corpus returns small distances of 0.04, 0.05, and 0.06 on MIMIC-III, PhysioNet, and Yahoo S5, with a maximum class-prior shift of 1.8 points, so selective gating reshapes *which* samples are auto-labeled without materially shifting the accepted-sample marginal.

F Calibration Features and Hyperparameters

The calibrator consumes an eight-feature vector $\phi_i \in \mathbb{R}^8$ summarising the 12-view ensemble: (1) *voting entropy* $H(\tilde{y}_i^{(1)}, \ldots, \tilde{y}_i^{(12)})$ over the empirical label distribution; (2) *agreement rate* as the majority-class share; (3) *max self-confidence* $\max_j c_i^{(j)}$; (4) *self-confidence std* across views; (5) *rationale similarity* as the mean pairwise cosine of MiniLM-L12 embeddings of $r_i^{(j)}$; (6) *evidence overlap* as the mean pairwise Jaccard similarity of cited change-point sets; (7) *temperature variance* as the within-temperature label dispersion averaged across the four sampling temperatures; (8) *prompt variance* as the within-prompt-template label dispersion averaged across the three prompt variants. The polynomial-degree-2 expansion adds 36 cross-terms; ridge regularisation $\lambda = 0.1$ is selected by 5-fold CV. Table 6 summarises hyperparameters.

G Prompt Template Schema

Each of the three prompt variants follows a fixed JSON schema that pairs the textual time-series summary with explicit output and rationale slots. The standard variant reads:

```
{ "context": "30-second window centred on alarm onset", "summary": {...statistical descriptors...}, "change_points": [...],
```

Table 6: CALM-TS hyperparameters used in all reported runs.

Component	Value	Notes
Prompt variants M	3	narrow / standard / wide context
Temperatures $\lvert\mathcal{T}\rvert$	4	$\{0.0, 0.3, 0.7, 1.0\}$
Views per sample m	12	$M \times \lvert\mathcal{T}\rvert$
Gold-set size n	500	per seed, per dataset
Calibrator	LogReg	degree-2 polynomial, $\lambda = 0.1$
CV folds	5	out-of-fold $\hat{p}_i$ for τ
Risk target α	0.05	default; 0.03/0.07/0.10 swept
Confidence δ	0.05	Theorem 1
Drift trigger	−15% Cov	200-window rolling baseline
Verification budget B	30%	of rejected set, knapsack

```
"ml_hint": "class_prob = {...}", "instruction":
"Classify the alarm as TRUE or FALSE; cite
at least one numeric feature; respond as JSON
{label, confidence, rationale, evidence_ids}." }
```

The narrow variant restricts the summary to a 15-second window and drops the cross-channel correlation block, forcing the LLM to reason on local morphology alone. The wide variant doubles the window to 60 seconds and prepends two retrieved nearest-neighbour cases. Temperature sweeps the four values $\{0.0, 0.3, 0.7, 1.0\}$ on every variant. Full templates including the system message, the JSON validation schema, and the per-dataset few-shot exemplars are listed in Suppl. §H.

H Reproducibility Artefacts

All PhysioNet numbers in Table 1 are reproducible from the released artefact directory `results/runs/seeds0_1_2_3_4/`, which aggregates the per-seed runs `seed0/`–`seed4/` (10 methods × 5 seeds; 500 gold and 1,475 test windows per seed) into a top-level `results.json` reporting risk, coverage, cost, and F1 with the corresponding standard errors. Per-cell traces are exposed under `cells/<method>__seed<i>.json` as relative symlinks back to the originating `seed<i>/` directory. Each seed directory carries a `manifest.json` that records git SHA, Python and package versions, dataset hashes, gateway URL, and LLM token usage, together with the audit log `raw_trace.jsonl.gz` (∼16 MB gzipped, 76,620 prompt–response pairs for seed 0) and a wall-clock progress log `run.log.gz`; SHA256 and byte count in the manifest permit offline integrity verification of both. The task–feature ablation in Table 2 additionally references `multi5class/` and `multi5class_clinical/`, which share backbone and split protocol with the main runs.

Backbone and gateway provenance. The PhysioNet run executes against an OpenAI-compatible HTTPS gateway, `OPENAI_BASE_URL = https://yunwu.ai/v1`, routing to the official gpt-4o-mini endpoint. Any OpenAI-compatible deployment of gpt-4o-mini at temperature 0.0 for the calibration sweep, paired with the per-view temperature schedule of Section 5, reproduces the reported numbers to within per-seed standard error. The seed-0 manifest records git SHA `521be6f8` with parameter-passing edits to `calm_ts_core.py`, `data_loaders.py`, and `run_main_real.py`. Seeds 1–4 advance to SHA `37102a18`; the only tracked-file edit across seeds 2–4 is a parameter-passing change to `calm_ts_core.py` outside the gating and calibration logic. The remaining `dirty` entries on both SHAs are untracked launch scripts and result directories; the exact `dirty_files` list and per-seed wall-clock are logged in each `manifest.json`.

Per-seed completeness audit. Seeds 0, 3, and 4 ran as complete 10-baseline sweeps (76,592–76,620 LLM calls, 9h46m–17h36m wall-clock each). Seeds 1 and 2 carry shorter manifests (1,475 and 3,245 calls) covering the post-hoc resume of LLMaAA (seed 1) and FreeAL + LLMaAA (seed 2); their remaining eight cells were inherited unchanged at the same code revision and dataset SHA256s.

Token usage. Per-seed token consumption is reported individually rather than summed. Seed 0 logs 73.2 M input and 9.1 M output tokens through gpt-4o-mini across 76,620 calls; the full 5-seed aggregate is 224.8 M input and 27.9 M output tokens. Per-seed breakdowns are exposed in `seed<i>/manifest.json`.

MIMIC-III and Yahoo S5 runs execute against GPT-4 under the same 5-seed gold-split protocol; the corresponding artefacts ship alongside this paper. Section-level proof sketches, the cost-model derivation, the baseline implementation roster, and the drift detector specification are inlined in Appendices A–D; the supplementary appendix holds the remaining material (extended ablation tables, calibration feature engineering, prompt templates, and the streaming case studies).

Code availability. The CALM-TS framework source and a CPU demo are released at https://github.com/qqyjx/CALM-TS; the complete pipeline (preprocessing, multi-view querying, calibration, gating, and evaluation), the figure-generation scripts, and the per-seed run artefacts are archived in the Zenodo deposit `10.5281/zenodo.20506015` (also cited in the Resource Availability statement on page 1).

Compute footprint and bit-exact caveat. The 5-seed PhysioNet sweep runs in 60–65 wall-clock hours on a single workstation, dominated by the multi-view labeling step; MIMIC-III and Yahoo S5 sweeps under GPT-4 complete in ∼24 hours per dataset. Per-seed numbers in Table 1 reproduce to within standard error on any OpenAI-compatible gpt-4o-mini gateway; bit-exact reproduction depends on the gateway's tokenisation and sampling.

Robustness checks. Three independent stress tests confirm the gold-set protocol. (i) *Gold-pool drift*: shuffling the gold and pool partitions across seeds keeps MIMIC-III risk in $[0.044, 0.051]$. (ii) *Calibrator family swap*: replacing logistic regression with XGBoost lifts AUC to 0.907 on MIMIC-III with identical risk control. (iii) *Prompt-template drop*: ablating any one of the three prompt variants raises risk by ≤ 0.004 and reduces coverage by ≤ 0.03, indicating no single template is load-bearing.

Ethics. MIMIC-III is used under the PhysioNet credentialed-access agreement; PhysioNet Challenge 2015 under ODC-By; Yahoo S5 under Yahoo Webscope. No personally identifiable information leaves the gateway. All clinical interpretations in the audit trail are scoped to research evaluation; production clinical deployment retains human verification on every accepted label as a regulatory matter.